



International College of Economics and
Finance

Big Data and Machine
Learning

Moscow 2025

Final submission

Employee Attrition Prediction

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Key Information

Taken from [Kaggle](#)

Its main aim is to support the development of predictive models for employee attrition, which is in our opinion a critical human resource management challenge.

14 900

obs. in the test set

59 598

obs. in the training set

47.55 %

“Attrition” equal to 1
(which means they left their jobs)

52.45 %

“Attrition” equal to 0
(which means they stayed)

Age

Years at Company

Monthly Income

Distance from Home

There are 4 continuous variables, while the others are discrete



Exploratory Data Analysis (EDA)

Continuous variables

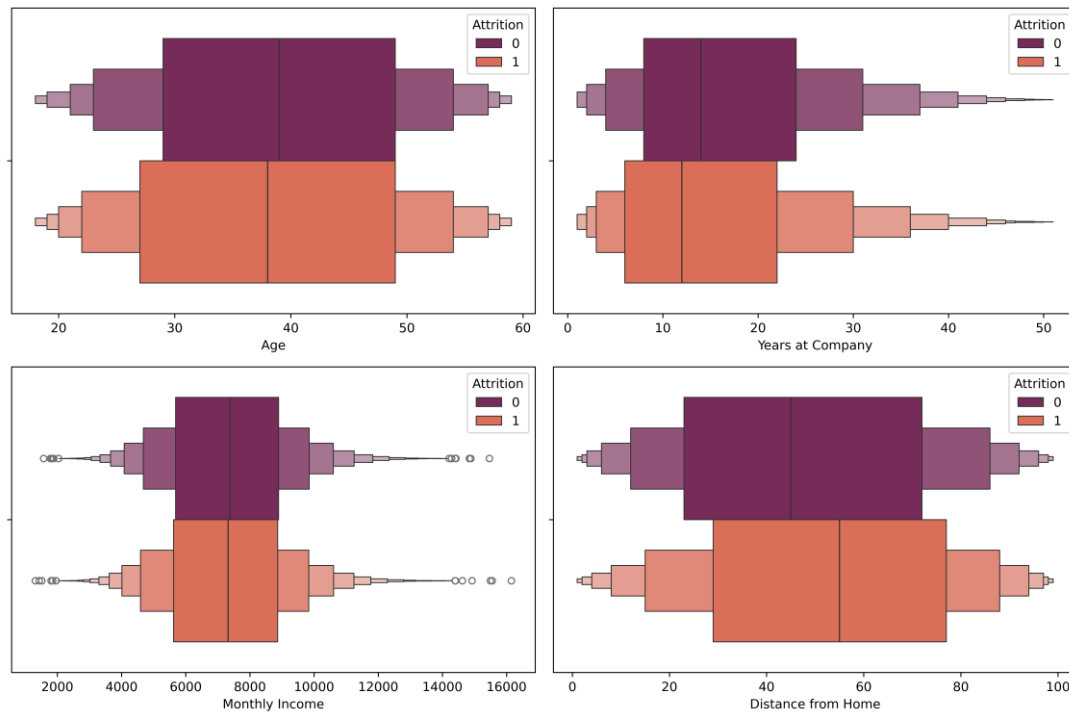


Fig. 1 Continuous variables distributions

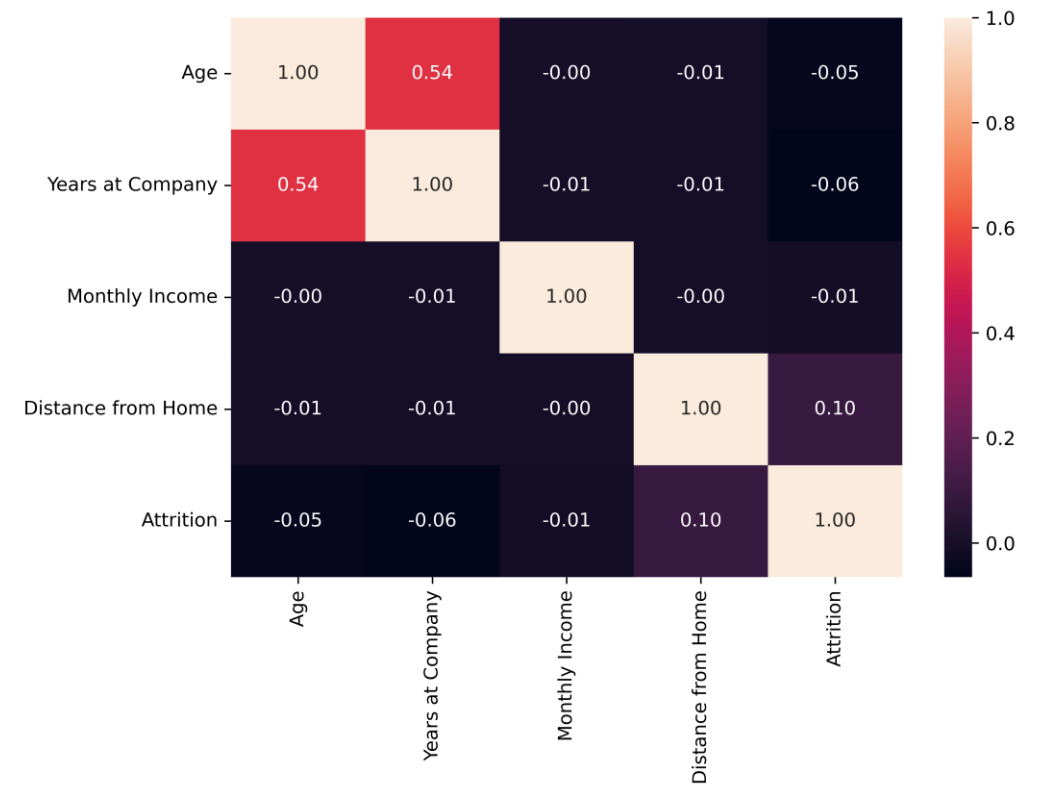


Fig. 2 Correlation matrix of continuous variables

Exploratory Data Analysis (EDA)

Discrete variables

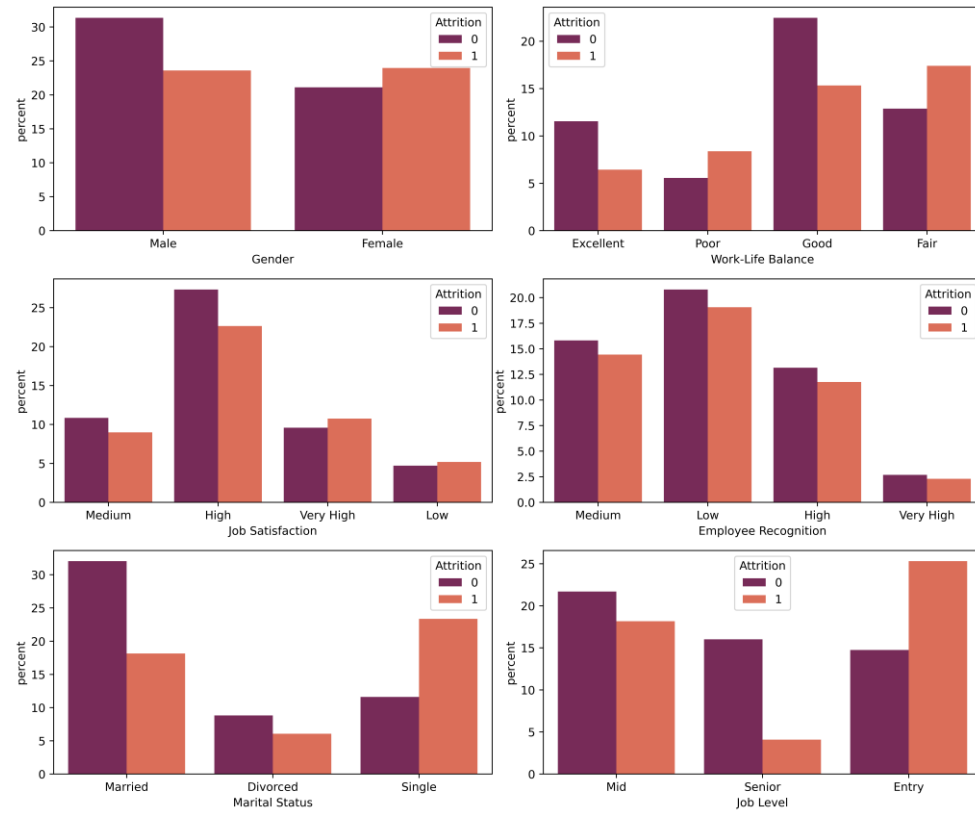


Fig. 3 Main discrete variables

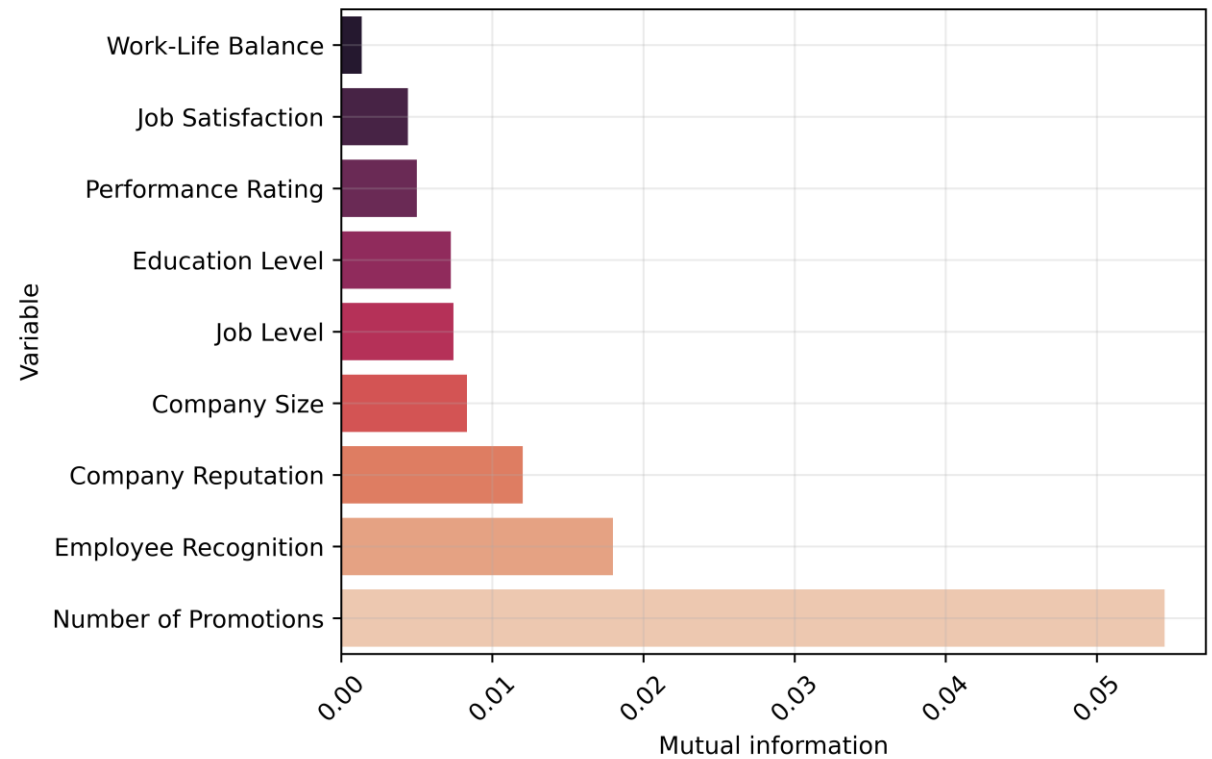


Fig. 4 Mutual information for ordinal variables



Modelling

Logistic regression

The default logistic model was trained on the train set with all initial variables plus new variables and one principal component for variables connected to age.

- In-sample recall is 0.74
- Recall on inference is 0.7369
- (Additionally) accuracy is 0.7369127

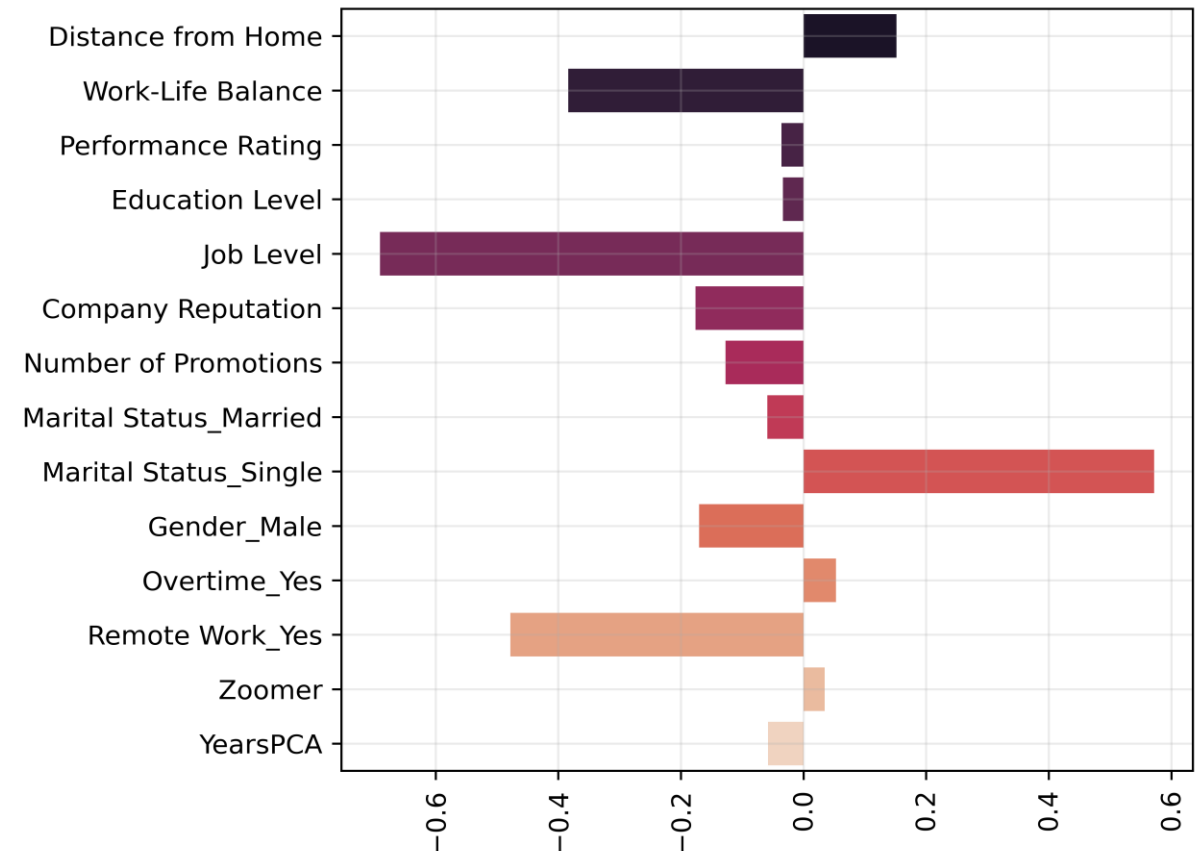


Fig. 7 Logistic regression feature importance

Modelling

Decision tree

Decision tree was trained on the same feature set as before. After grid search the results are the following:

- Recall on inference is 0.749
- Accuracy on inference is 0.736

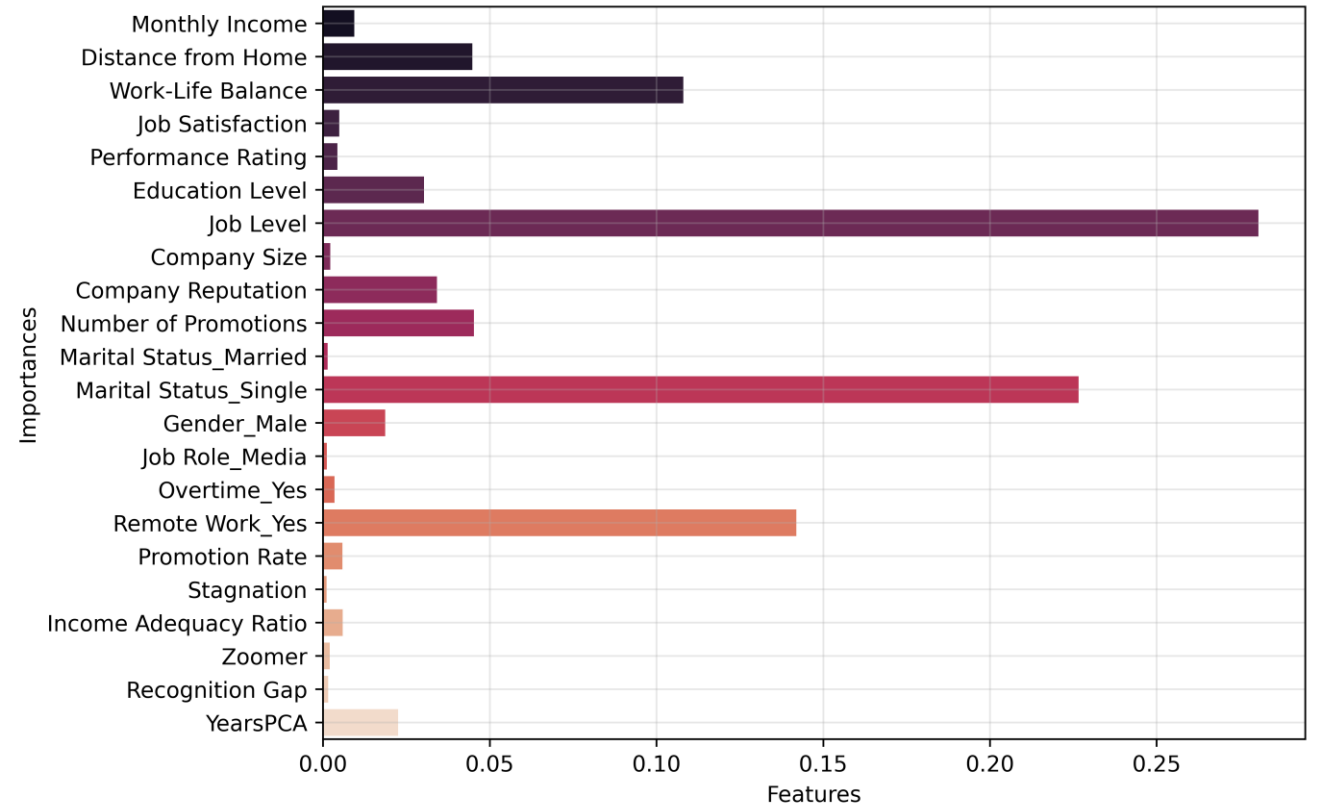


Fig. 8 Decision tree feature importance



Ensemble Methods

- We trained several ensemble models (applying Optuna optimization algorithm to get the hyperparameters) – RandomForest, XGBoost, CatBoost and implemented self-made voting estimator for three sets of models – (logistic regression, decision tree, random forest), (logistic regression, decision tree, xgboost) and (logistic regression, decision tree, catboost) in order to account for linearity and non-linearity simultaneously.
- However, the results on inference turned out to be only slightly better than for the models before.
- Moreover, for the random forest and boostings same features as before appeared to be important while the least important were “Job Role”, “Stagnation”, both types of opportunities and “Employee Recognition”.



Validation and discussion

- All three models (forest and boostings) appeared to be robust on both train and test sets. On the train set, random forest has mean recall equal to 0.72 (+- 0.006 std), xgboost - 0.74 (+- 0.005 std) and catboost – 0.742 (+- 0.007 std). On the test set the results for recall are the following: 0.71 (+- 0.017 std) for forest, 0.732 (+- 0.016 std) for xgboost and 0.733 (+- 0.013 std) for catboost. For accuracy scores the situation is similar.
- Hence, the most important insight we got – not always complex and non-linear models can outperform basic (even though with regularization) linear or simple non-linear models.
- For the purpose of the project and its empirical implementation we would recommend to use a regularized decision tree or logistic regression since they are easily interpretable and perform well comparing to the black box models we had.

For more information kindly address the attached report file.

